

Formula Zynerji

A Blockchain-Based, Mechanism-Design Redesign of Formula 1 for Meritocracy

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Abstract

Formula Zynerji re-derives the Formula 1 rulebook from a single objective — **meritocracy** — using game theory and seventy years of how teams have actually behaved. Its spine is a **blockchain**: every part that runs on a car must have its full design (CAD/FEA/CFD) uploaded to an immutable ledger, and only on-chain parts are legal. Forced disclosure converts R&D from a private secret into a shared club good, which collapses the spending war and lets the field *self-balance by copying* — with no development tokens, no auction, and no Balance of Performance. A mandatory minimum-spend floor forces continuous innovation; the resulting shared, race-validated engineering corpus is sold to outside industry to fund the sport and the prize money. The car is a deliberate kitbash of historical F1 eras around a distinctive 2.5 L two-stroke uniflow diesel. This paper presents the thesis, the mechanisms (catalogued M1–M13), the car, the competition format, and the economic model that pins the two keystone numbers.

1 Thesis: the championship as a measurement instrument

A season is an *estimator* of an unobservable quantity — true competitive merit (car + team + driver). The standings are the estimate; meritocracy is the program of minimising the gap between them:

$$\text{Standings} = \text{Merit} + \text{Luck} + \text{Budget-bias} + \text{Gaming-bias}.$$

Every rule is an intervention on one of those error terms, and is judged by one test: does it make the finishing order track merit more closely? Formula 1's instinct has been to remove danger and cost by *banning* innovations (flat floors 1983, the turbo ban 1989, the electronic-aid ban 1994). Formula Zynerji keeps the engineering freedom of the 1966–1982 open era but contains its two killers — danger and runaway cost — with a tight *box* (a non-negotiable safety floor, a single energy/power pinch-point, and a self-balancing economy) rather than prescriptive bans. The design razors and the full mechanism catalogue (M1–M13) live in the project's *mechanism-design* document.

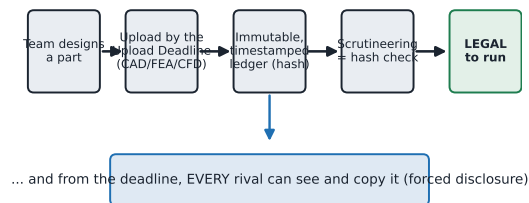
2 The spine: forced on-chain disclosure (M10/M11)

Every run part's complete design record is committed to an immutable, timestamped chain by the start of each event; **only on-chain parts are legal**, so scrutineering reduces to a hash check and a secret or back-dated part is structurally impossible. The economic consequence is the point: an innovation, once raced, is visible and copyable, so the value of any one advantage is capped at the head-start it buys. No team will spend millions for an edge that arms its rivals within a weekend — equilibrium R&D spend collapses, and the contest shifts from *hiding the biggest secret* to *innovating fastest and integrating best*. The innovator's reward is the transient

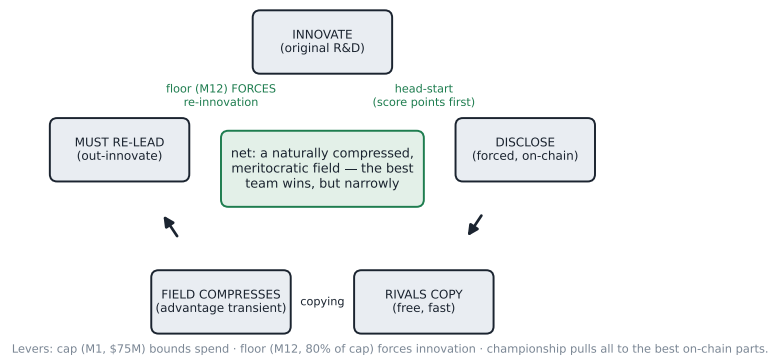
head-start (you are fast first, and score points, before rivals can manufacture a copy) plus reputational recognition on a published Innovation Index. There are no tokens and no bounty.

FORMULA ZYNERJI — BLOCKCHAIN SPINE, SELF-BALANCING ECONOMY & REVENUE

A. ON-CHAIN LEGALITY (only on-chain parts may run — M10)



B. THE SELF-BALANCING ECONOMY (no tokens, no auction, no handicap)



C. REVENUE — the shared R&D corpus is a product that funds the sport

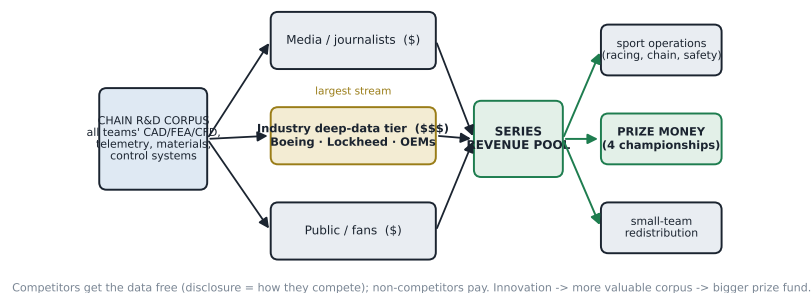


Figure 1: The blockchain spine: on-chain legality (A), the self-balancing economy (B), and the revenue streams that fund the sport (C).

3 The self-balancing economy and its revenue

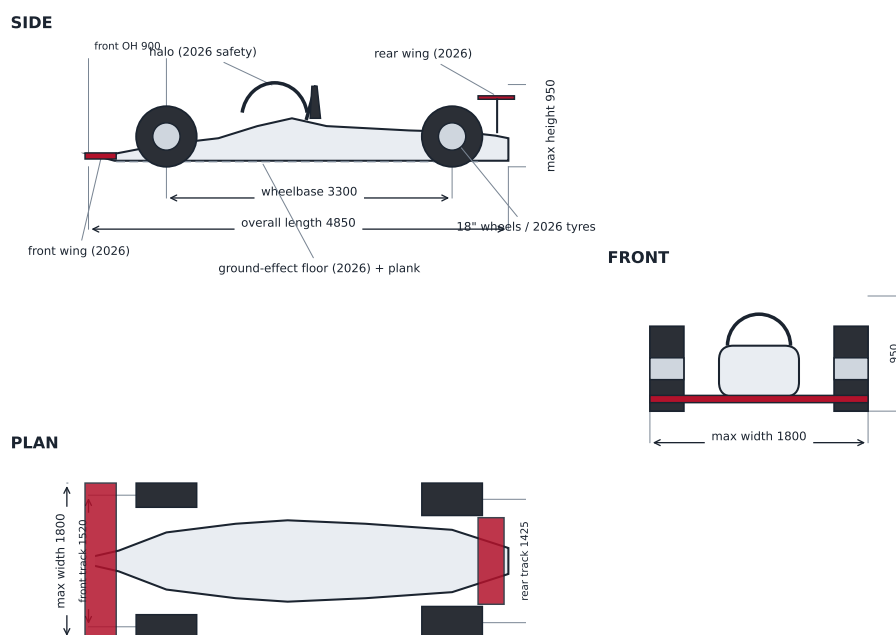
Because forced disclosure makes copying free, the field balances itself: any advantage is copied away (the field compresses), the championship pulls every team toward the best on-chain parts, and re-leading demands fresh innovation. This removed the need for the artificial balancing machinery of earlier drafts — development tokens, a development auction, and a success handicap are all **deleted**. The only economic levers are a hard, money-neutral **cap** and a mandatory minimum-spend **floor**. The floor is the keystone: a token-free copy economy has exactly one bad equilibrium — everyone copies, nobody innovates — and the floor forecloses it by forcing every team to do original development.

The shared corpus is also a **product**. Forced disclosure makes the chain a continuously-growing, cross-validated, race-proven engineering dataset, sold through tiered access: competitors get it free (disclosure is how they compete); journalists pay a media tier; and **non-competing industry** — aerospace, defence, automotive and energy firms such as Boeing and Lockheed — license the deep CAD/FEA/CFD, materials and control data for their own R&D at a premium. This is the largest stream, and it funds the series’ operations, the **prize money** for the four championships, and a redistribution toward smaller teams. Richer innovation makes the corpus more valuable, which grows the prize fund — a further reason to innovate.

4 The car: an era kitbash

The car deliberately ports specific historical F1 regulation sets: **2026** wings, floor, tyres and safety; **2021** hydraulically-interconnected suspension; a **2013** 7-speed gearbox; and **2008** chassis dimensions and body rules. A 3300 mm wheelbase, 1800 mm width and ~605 kg minimum mass give a planted but ferociously powerful machine. Aero is disciplined not by ever-more-prescriptive geometry but by *dirty-air externality pricing* (M4): each car’s wake is measured, and a clean-wake car is granted a larger design-stage downforce allowance — so close racing and overtaking happen on merit, and there is no DRS.

FORMULA ZYNERJI — CAR GENERAL ARRANGEMENT (era-kitbash: 2026 aero/tyres/safety • 2008 dims • 3300 mm WB)



All dimensions in mm. First-pass GA from the dimensional set (design/chassis-integration.md).

Figure 2: Car general arrangement (side / plan / front) with the dimension set.

5 The powertrain: a 2.5 L two-stroke uniflow diesel

The engine is a 2.5 litre inline-five, two-stroke, uniflow-scavenged compression-ignition (diesel) unit on a JP-8-class synthetic e-kerosene, force-fed by a mechanical screw (Lysholm) super-

FORMULA ZYNERJI — PARAMETRIC 3D GENERAL ARRANGEMENT (CadQuery → STEP)
ISOMETRIC

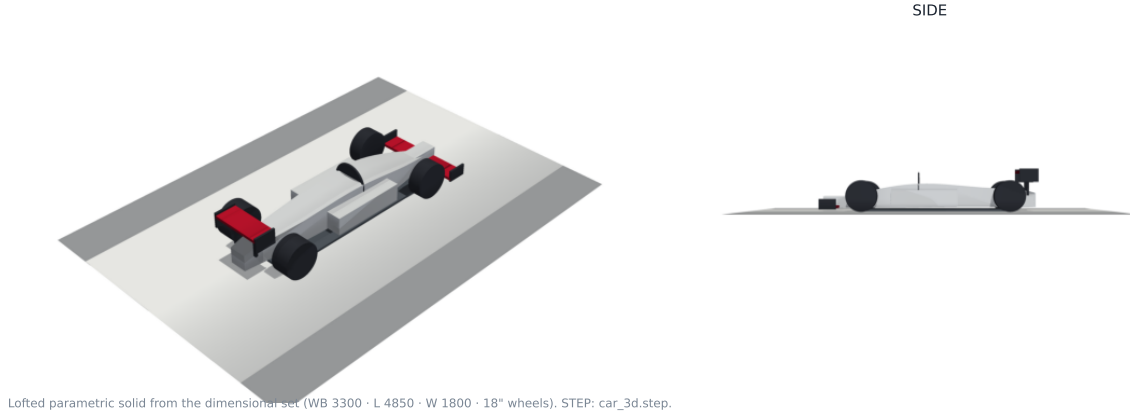


Figure 3: Parametric 3D general arrangement (CadQuery), exported to STEP.

charger, with no traction hybrid. Uniflow scavenging — fresh charge in through liner ports at the bottom, exhaust out through poppet valves in the head — is the most efficient two-stroke method and underpins the high BMEP. At 26 bar BMEP and 7000rpm it makes ~ 758 kW (~ 1015 hp) and ~ 1035 N m; at the 605 kg minimum that is ~ 1.25 kW/kg, turbo-era-F1 power-to-weight. Exhaust tuning is by an electronically-controlled variable-length manifold (set-and-run): an automated *engine* optimisation, not a driving aid.

FORMULA ZYNERJI — 2.5 L INLINE-5 TWO-STROKE UNIFLOW DIESEL

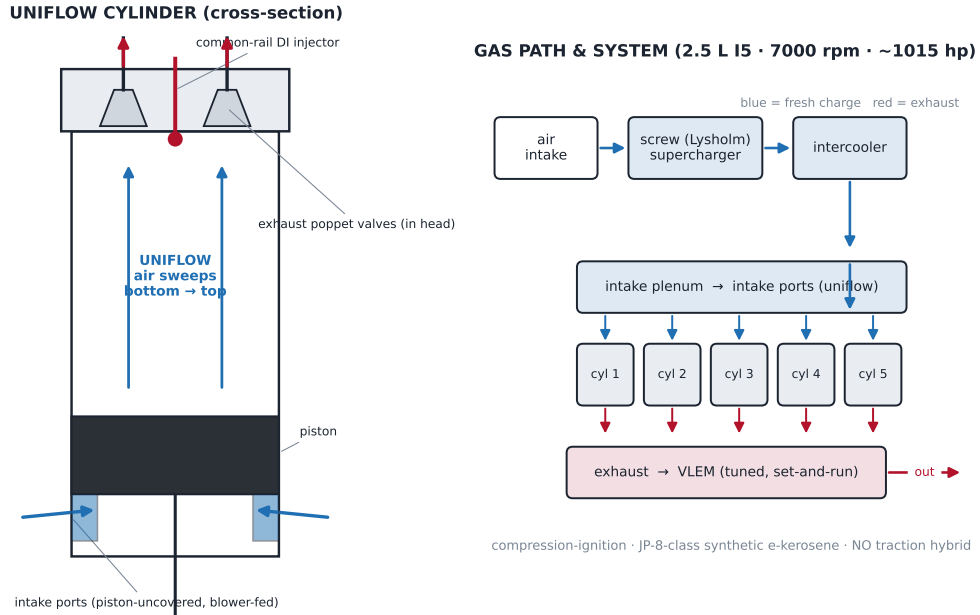


Figure 4: The 2.5 L I5 two-stroke uniflow diesel: cylinder cross-section and gas-path.

6 Driver-skill primacy and the vectoring differential

The governing electronics principle is *automate the engine, never the driving*: live manual driver inputs are encouraged; automated engine-output optimisation is permitted but locked before the race; automated car-control aids (traction control, ABS, closed-loop vectoring) are banned.

The rear differential is a **driver-vector**ed twin wet-clutch unit (GKN Twinstar-type): two steering-wheel triggers set the percentage lockup of each side’s clutch in real time. It is legal precisely because it is *open-loop* — the trigger-to-clamp map is fixed, declared and on-chain, and the safety ECU adds no sensor feedback — which is what distinguishes it from banned automated vectoring.

FORMULA ZYNERJI — DRIVER-VECTORED REAR DIFFERENTIAL (twin wet-clutch, GKN Twinstar-type)

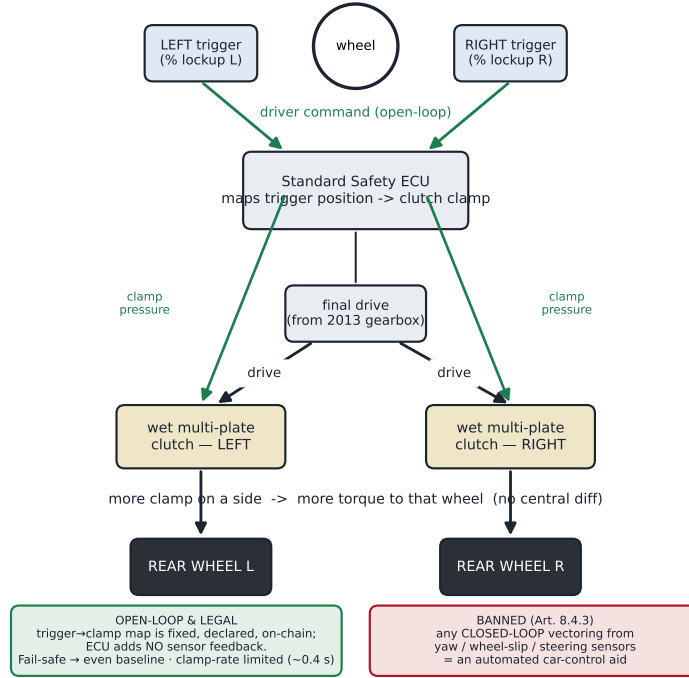


Figure 5: The driver-vectoring twin wet-clutch differential and its open-loop control law.

7 The competition: a merit estimator

The competition is tuned for signal-to-noise. The points curve scores *all* classified finishers on a convex 40→1 scale, so mid-grid merit is measured rather than rounded to zero. The Drivers’ title is complemented by an advisory **Driver-Merit Index**: a hierarchical Bayesian model that separates a car effect from a driver effect using teammate comparisons (identical machinery), tied onto one scale and reproducible from on-chain data — revealing the best driver net of machinery without corrupting team behaviour. A signature rule replaces blue flags with *lapping elimination*: a backmarker may fight, but if it is lapped by the leader it is black-flagged out, and the leader scores a point. Because only the leader can ever lap a car first, the rule cleanly removes lapped-traffic interference — every car still running is on the lead lap.

8 Safety: the invariant

Safety is the one part of the formula that is never an objective to be optimised and never traded: a non-negotiable, outcome-based floor (crash-test pass criteria, a halo-equivalent cockpit structure, homologated energy storage, driver equipment, and circuit/medical/spectator-separation minima written at the ruleset level). Engineering freedom and rigid safety standards are orthogonal — one can run a tightly tuned competition while mandating exact crash-test outcomes.

9 Economic model: pinning the cap and the floor

The two keystone numbers were set from an explicit model rather than guessed. **Cap.** A bottom-up cost build for a lean, shared-design program puts a competent entry at $\sim \$67\text{M}$ and a running-cost floor at $\sim \$56\text{M}$, so a **\\$75 M** cap is $\sim 1.1\times$ competent — a genuinely tight constraint with real development headroom, while still binding the teams who would outspend it. **Floor.** An innovate-versus-copy game shows that with no floor, teams self-select only $\sim 13\%$ original-R&D (the public-goods under-provision that justifies the floor). Vitality rises monotonically with the floor at no cost to merit-correlation or field-compression, so the floor is a pure vitality-versus-affordability lever; and because the running-cost floor is $\sim 75\%$ of the cap, the floor must sit above it to bite — hence **80% of the cap**.

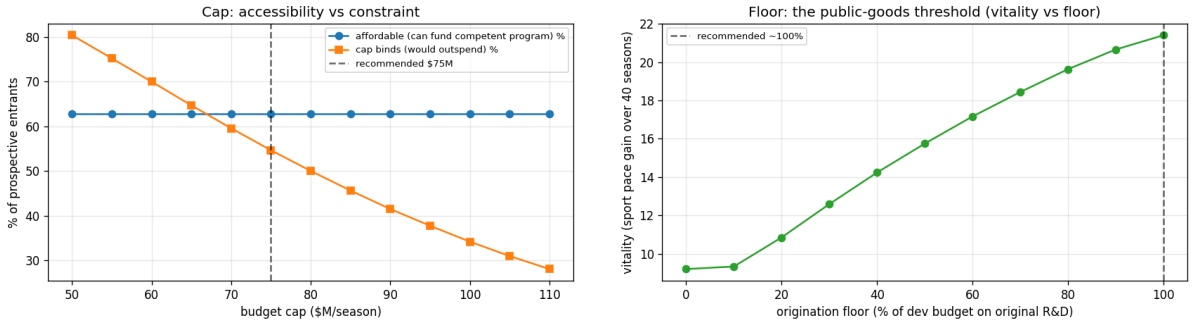


Figure 6: Left: cap accessibility versus constraint. Right: the floor as the public-goods threshold (vitality versus the origination floor).

10 Conclusion

Formula Zynerji is, in one line, the engineering freedom of 1970s Formula 1 wearing 2020s safety and a blockchain economy. By forcing disclosure, capping money, mandating a contribution floor, pricing the dirty-air externality, keeping the driving manual, and measuring driver merit explicitly, it makes the championship a faithful instrument for merit — and turns the sport’s own shared R&D into the revenue that funds it.

The full regulations (technical, sporting, safety, financial), the mechanism catalogue, the economic model, and the figure sources are openly available at github.com/Zynerji/FormulaZynerji.

A Mechanism catalogue (M1–M13)

Every rule traces to a mechanism. The nine design razors (R1–R9) that generate them — e.g. *equalise inputs, never outputs; price externalities, don’t prescribe around them; design for the equilibrium, assume Goodhart* — are set out in the project’s **mechanism-design** document. “Removed” entries are retained for the record: they were superseded either by the self-balancing economy or by the move to conventional racing, and the table documents *why* each was dropped.

ID	Mechanism	Role	Status
M1	Money-neutral hard cap	Equal, hard \$75 M budget cap; no luxury-tax lane. Removes budget-bias.	Active
M2	Development-resource auction	Token market rationing wind-tunnel/CFD/dyno throughput.	Removed (self-balance)
M3	Merit-weighted handi-cap	Success-scaled cut to a team’s <i>future</i> development.	Removed (self-balance)
M4	Dirty-air externality pricing	Wake is measured; a clean-wake car earns a larger design-stage downforce allowance. Enables merit overtaking → no DRS.	Active
M5	Luck-suppressing neutralisation	Gap-preserving / time-neutral safety-car correction.	Removed (conventional racing)
M6	Championship-estimator design	All-finishers convex points curve (40→1) tuned for signal-to-noise.	Active
M7	Driver-merit disentanglement	Advisory Driver-Merit Index: a Bayesian car-vs-driver split from teammate pairs.	Active
M8	Anti-gaming sporting rules	Sensor-measured track limits; broad parc-fermé; scrutineering by hash.	Active
M9	Robust cost monitoring	Related-party pricing, randomised audits.	Subsumed by M10
M10	Forced on-chain disclosure	Only on-chain parts are legal; the spine. Collapses spend; advantages go transient.	Active (spine)
M11	Innovation incentive	The reward to innovate: transient head-start + Innovation-Index recognition (no tokens).	Active
M12	Minimum-spend floor	Mandatory ≥80% of cap on development, ≥half original on-chain work; forecloses copy-only stagnation.	Active (keystone)
M13	Lapping elimination	No blue flags; lapped by the leader → black-flagged out, leader +1. Removes lapped-traffic luck.	Active

The active set is therefore: the blockchain spine (M10) and its two innovation safeguards (M11 carrot, M12 stick); a money cap (M1); dirty-air pricing (M4); and the competition estimators and rules (M6–M8, M13). The economy needs no artificial currency (M2/M3 gone), and output-side luck is left to conventional racing (M5 gone).

B Dimension & specification set

First-pass values; the era-kitbash basis and the derivation are in the project’s **chassis-integration** document, the engine and systems in **technical.md**, and the cap/floor in **economic-modeling**.

	System	Era basis
B.1 Era basis	Front & rear wings, floor, tyres, wheels, safety	2026
	Suspension (hydraulically interconnected)	2021
	Gearbox (7-speed + reverse)	2013
	Chassis dimensions & body rules	2008
	Engine, vectoring diff, economy	new

B.2 Dimensions & mass	
Parameter	Value
Wheelbase	3300 mm
Overall length	≈ 4850 mm
Max width	1800 mm
Max height	950 mm
Front / rear track	≈ 1520 / 1425 mm
Wheels	18 in
Tyre dia. F / R	≈ 705 / 690 mm
Min. mass (car+driver, no fuel)	605 kg
Race fuel	≈ 80 kg / 100 L
Floor	2026 ground-effect, $\approx 0.92 \times$ scaled

B.3 Powertrain & economy	
Parameter	Value
Engine	2.5 L I5 2-stroke uni-flow diesel
Induction	screw (Lysholm) supercharger
BMEP / rev ceiling	26 bar / 7000 rpm
Peak power	≈ 758 kW (≈ 1015 hp)
Peak torque	≈ 1035 N m
Power-to-weight	≈ 1.25 kW/kg
Injection	common-rail DI, ≈ 2500 – 3000 bar
Exhaust	electronic VLEM (set-and-run)
Fuel	JP-8-class synthetic e-kerosene
Budget cap / floor	\$75 M / 80% of cap